

The Valuation Reading of the Qubit Delusion: Is the Adelic Kernel Load-Bearing?

Rowan Brad Quni-Gudzinis

2026-08-01

Author: Rowan Brad Quni-Gudzinis | **Date:** 2026-08-01 | **License:** QNFO-ULA: <https://legal.qnfo.org/>

The Valuation Reading of the Qubit Delusion

Abstract

The Qubit Delusion (DOI 10.5281/zenodo.21254143) argues that the qubit-gate-circuit model of quantum computation imports a particle ontology inconsistent with quantum field theory, and that the field’s failure to deliver commercially viable hardware after \$35 billion of investment is an epistemic crisis, not merely an engineering delay. The Adelic Core Research Program (ACRP) has developed a valuation-theoretic framework — Ostrowski’s theorem, p-adic valuations, Bruhat-Tits trees, adèles — as a candidate mathematical substrate for quantum computation. The K3 audit (ACRP-01, DOI 10.5281/zenodo.21727314) found that the Qubit Delusion makes NO explicit use of this kernel. This paper asks whether the bridge between the two programs is load-bearing: does valuation theory constrain, sharpen, or bound the Qubit Delusion’s claims? We examine the three candidate junctions — the particle-ontology critique, the error-correction overhead wall, and the substrate-is-algorithm thesis — and find that the kernel is aesthetic, not load-bearing, at every junction: the Qubit Delusion’s claims are already correct (or incorrect) without the kernel, and the kernel’s contribution is a descriptive re-framing, not a novel bound. We argue that this is the honest and publishable result, and we register calibration predictions for when a genuine load-bearing junction would be recognized.

1 Introduction

The Qubit Delusion [1] is a systematic critique of the qubit-gate-circuit model of quantum computation. Its central thesis is that the model imports a particle ontology — qubits as localized, particle-like entities — that is inconsistent with relativistic quantum field theory, relational quantum mechanics, and the physics of continuous, correlated systems. The paper argues that the field’s failure to deliver commercially viable hardware is therefore not primarily an engineering problem but an epistemic one: a map-territory confusion operating at industrial scale.

The Adelic Core Research Program (ACRP) offers a different diagnosis of the same landscape. Its central claim is that non-Archimedean mathematics — p-adic valuations v_p , Bruhat-Tits trees, Ostrowski’s theorem, and the adèle ring — provides the correct state-space geometry for fundamental physics and quantum computation [2]. The program has developed a “kernel membership” audit

that classifies research programs by their use of four kernel elements: valuation, tree boundary, Ostrowski, adeles.

The K3 audit within ACRP-01 [2] examined the Qubit Delusion and found that it makes NO explicit use of this kernel. The Qubit Delusion’s critique stands entirely on Archimedean-accessible reasoning: particle ontology is inconsistent with QFT, error correction is expensive, and the field has delivered no hardware.

This paper asks the question that the K3 audit leaves open: is the absence of kernel use a gap to be filled, or is it evidence that the kernel is irrelevant to the Qubit Delusion’s claims? We examine three candidate junctions where the kernel might be load-bearing.

2 The K3 Audit Finding and the Kernel Membership Framework

The ACRP-01 consilient synthesis [2] organizes the QNFO research program around a kernel-first spine: valuation $\rightarrow$ tree boundary $\rightarrow$ Ostrowski $\rightarrow$ adeles. The kernel membership audit assigns each pillar of the program a membership grade: FULL (uses all four elements), WEAKER (uses a subset), or NONE.

The Qubit Delusion received NONE. This is notable because the Qubit Delusion and the ACRP program are making related claims about the future of quantum computation: the Qubit Delusion says the dominant model is epistemically broken; the ACRP program says the correct model is non-Archimedean. If both claims are true, there should be a junction where the non-Archimedean diagnosis sharpens the epistemic critique.

The K3 audit finding can be read two ways: 1. **As a gap:** the Qubit Delusion missed an opportunity to connect its critique to the non-Archimedean framework. 2. **As a signal:** the Qubit Delusion’s critique does not need the kernel — it stands or falls on its own terms, and the kernel’s absence is evidence that the two programs are orthogonal.

This paper investigates which reading is correct.

3 Junction 1: The Particle-Ontology Critique

3.1 The Qubit Delusion’s Claim

The Qubit Delusion argues that the qubit-gate-circuit model treats qubits as localized particle-like entities, and that this ontology is inconsistent with quantum field theory, where particles are emergent phenomena and fields are fundamental. The critique is epistemic: the model’s conceptual architecture misrepresents quantum reality, and this misrepresentation has engineering consequences.

3.2 The Kernel’s Candidate Contribution

The valuation-theoretic framework offers a specific mathematical characterization of what a qubit is: a qubit is a two-state system, and the set of two-element subsets of a p -adic digit set carries a natural ultrametric structure. More substantively, the Bruhat-Tits tree for $SL(2, \mathbb{Q}_p)$ is a $(p + 1)$ -regular tree whose vertices are homothety classes of lattices — a natural state space for a two-level system embedded in a p -adic field.

The candidate load-bearing claim would be: the particle ontology of the qubit-gate-circuit model fails because it uses Archimedean (real) state-space geometry, and the correct geometry is the

ultrametric tree. This would connect the Qubit Delusion’s epistemic critique directly to the ACRP kernel.

3.3 Why It Is Not Load-Bearing

The Qubit Delusion’s critique does not depend on the specific geometry of state space. The critique identifies four scaffolds — qubit-as-particle, gate-as-discrete-operation, decoherence-as-enemy, error-correction-as-classical-coding — and argues that all four are arbitrary conventions that misrepresent quantum reality. The argument against qubit-as-particle is that particles are not fundamental in QFT; it does not claim that the correct replacement is a tree geometry. The critique is agnostic between Archimedean and non-Archimedean replacements — it would be equally valid if the correct replacement were a continuous-variable field-theoretic model (the Qubit Delusion’s own preferred direction [3]).

In valuation-theoretic terms: the Qubit Delusion’s critique identifies a mismatch between the model’s ontology and quantum reality, but it does not constrain which alternative ontology is correct. The kernel provides one candidate alternative (ultrametric tree geometry), but the critique would survive unchanged if the kernel were false. **The kernel is not load-bearing at this junction** — it is a possible destination for a journey the critique does not require.

4 Junction 2: The Error-Correction Overhead Wall

4.1 The Qubit Delusion’s Claim

The Qubit Delusion cites the error-correction overhead wall as empirical evidence for its thesis: surface codes require thousands of physical qubits to encode a single logical qubit, fault-tolerant machines are projected for the 2030s, and the field has absorbed \$35 billion with zero commercially viable machines.

4.2 The Kernel’s Candidate Contribution

The ACRP program proposes that ultrametric error-correcting codes — codes whose geometry is native to Bruhat-Tits trees — could reduce this overhead [2, 4]. The candidate load-bearing claim: the overhead wall is an Archimedean artifact, and a valuation-theoretic code construction would break it.

4.3 Why It Is Not Load-Bearing

The overhead wall is a fact about the surface code, not about Archimedean geometry. The surface code’s overhead — $O(d^2)$ physical qubits for distance d , with logical error rate decreasing exponentially in d — is a property of the stabilizer formalism, not of the real numbers. The Qubit Delusion’s use of the overhead wall is as evidence that the qubit-gate model is expensive, not as evidence about geometry.

Moreover, the ACRP’s own program is honest about this: the paradigm forecast (ACRP-08, DOI 10.5281/zenodo.21747228) registers ultrametric QEC codes as its most promising candidate with a calibrated probability of only 0.12–0.25 by 2036, anchored to the historical base rate that zero mathematical framework adoptions have succeeded within a decade [5]. The kernel’s contribution to the overhead wall is a *possibility* (a tree-based code with surface-code-comparable thresholds), not a *bound* (a proof that Archimedean codes must fail).

The distinction matters: a load-bearing bridge would provide a novel constraint — e.g., a valuation-theoretic bound showing that Archimedean codes cannot achieve a certain threshold, or that ultrametric codes are provably superior. No such bound exists. The ACRP’s own forecast registers the threshold-comparability claim as an unbuilt construction. **The kernel is not load-bearing at this junction** — it is a research program with a speculative promise, and the Qubit Delusion’s empirical evidence stands independently of it.

5 Junction 3: The Substrate-Is-Algorithm Thesis

5.1 The Claim

“Beyond the Qubit” [3] argues that computation IS a physical process and the substrate IS the algorithm — pointing toward thermodynamic, neuromorphic, and optical architectures where physics does the computing directly. This is the Qubit Delusion series’ constructive direction.

5.2 The Kernel’s Candidate Contribution

The ACRP program’s strongest claim is that the substrate of computation is non-Archimedean: the adelic product formula constrains which computations are physically realizable, and the ultrametric tree is the natural state space for hierarchical computation [2].

5.3 Why It Is Not Load-Bearing

The substrate-is-algorithm thesis is a general philosophical claim: the physical implementation of a computation is not separable from the computation itself. This claim is compatible with any specific substrate hypothesis, including the non-Archimedean one — but it does not require it. A thermodynamic computer, a neuromorphic chip, or an optical interferometer are all instances of “physics does the computing directly,” and none of them require p -adic geometry.

The valuation-theoretic program can be read as a *specification* of the substrate-is-algorithm thesis (the substrate is the ultrametric tree), but the thesis is the broader claim and does not depend on the specification. **The kernel is not load-bearing at this junction** — it is a candidate instantiation of a thesis that does not require it.

6 The Honest Verdict

At all three candidate junctions — the particle-ontology critique, the error-correction overhead wall, and the substrate-is-algorithm thesis — the valuation kernel is aesthetic, not load-bearing. The Qubit Delusion’s claims are correct (or incorrect) on their own terms; the kernel provides a descriptive re-framing that neither strengthens nor weakens them.

The K3 audit’s finding that the Qubit Delusion makes NO explicit kernel use is therefore not a gap to be filled but a signal: the two programs are orthogonal. The Qubit Delusion is an epistemic critique of a specific engineering model; the ACRP program is a constructive proposal for a different mathematical substrate. They can coexist without a bridge, and attempts to force a bridge produce rhetoric, not results.

This is the honest and publishable result. The bridge is aesthetic.

7 Calibration Register

We register the following dated, strength-weighted predictions.

[**CHECK: 2027-01-01**] **No junction forced.** No peer-reviewed publication will claim that the adelic valuation kernel provides a novel bound on quantum error-correction overhead (a bound not obtainable from stabilizer theory alone) that is verified by independent computation. **Strength:** [WEAK] — anchored to the ACRP-08 forecast’s own calibration, which registers ultrametric QEC threshold-comparability as an unbuilt construction with probability 0.12–0.25 by 2036 [5]. **Status:** [PENDING]. **Risk:** “The bound was always implicit in stabilizer theory; the valuation-theoretic framing merely made it visible.”

[**CHECK: 2030-12**] **If the bridge becomes load-bearing, it will be via a threshold.** The only junction that could become load-bearing is Junction 2: a Bruhat-Tits tree code that matches or exceeds the surface-code threshold under identical noise models would constitute a novel, load-bearing result. **Strength:** [WEAK] — this is a conditional prediction: if a load-bearing junction emerges, it will be this one. **Status:** [PENDING]. **Risk:** “The forecast hedged by naming the most likely junction, which is not the same as predicting it will occur.”

[**CHECK: 2036-12**] **The Qubit Delusion’s critique stands independent of the kernel.** The Qubit Delusion’s epistemic critique (particle ontology is inconsistent with QFT; the field has delivered no hardware after \$35 billion) will remain valid regardless of whether ultrametric QEC codes succeed or fail. **Strength:** [STRONG] — the critique’s validity depends only on QFT and the empirical record, both of which are independent of the ACRP program. **Status:** [PENDING]. **Risk:** “This is a tautology — the critique was designed to be independent.”

[**CHECK: 2027-01-01**] **Null: no kernel-forced revision.** No revision of the Qubit Delusion will be published that attributes any of its conclusions to valuation-theoretic reasoning. **Strength:** [STRONG] — the K3 audit’s finding that the Qubit Delusion makes no kernel use is a fact about the published text, and no mechanism has been identified by which the kernel would force a revision. **Status:** [PENDING]. **Risk:** “A future revision might cite the ACRP program for rhetorical support without the kernel doing any work.”

8 Practical Applications

The honest verdict has a practical consequence: research effort should not be spent building rhetorical bridges between the Qubit Delusion and the ACRP program. The two programs have distinct value:

1. **The Qubit Delusion’s value** is as an epistemic cautionary tale and a filter: it identifies which quantum-computing claims are scaffold and which are invariant. Its reproducibility scorecard methodology [1] is directly applicable to evaluating quantum-computing claims.
2. **The ACRP program’s value** is as a constructive alternative: it proposes a specific mathematical substrate and registers falsifiable predictions (ACRP-08’s nine calibration entries [5]) that can be evaluated on their own terms.

The practical recommendation is separation: let the critique be a critique, and let the constructive program be a constructive program. Effort spent forcing a connection is effort not spent building ultrametric QEC codes or improving the reproducibility scorecard.

9 Conclusion

The K3 audit asked whether the Qubit Delusion’s absence of kernel use is a gap or a signal. We have argued it is a signal. At every candidate junction, the valuation kernel is aesthetic, not load-bearing: the particle-ontology critique does not require tree geometry, the error-correction overhead wall is a fact about stabilizer codes rather than about Archimedean geometry, and the substrate-is-algorithm thesis is compatible with any substrate hypothesis.

This is the honest and publishable result. The bridge is aesthetic. The Qubit Delusion and the Adelic Core Research Program are orthogonal programs with distinct value, and forcing a load-bearing connection between them would be an instance of the very map-territory confusion the Qubit Delusion diagnoses.

We register four calibration predictions: the strongest claims are that the Qubit Delusion’s critique stands independent of the kernel [STRONG] and that no revision will attribute its conclusions to valuation-theoretic reasoning [STRONG]. The conditional hope — that a Bruhat-Tits tree code could make Junction 2 load-bearing — is registered honestly as a weak, speculative possibility consistent with the ACRP program’s own calibrated forecast.

Declarations

Funding: This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Conflicts of Interest: The author is the sole contributor to the Adelic Core Research Program (ACRP), which produced the papers examined in this assessment. The author has no financial or institutional conflicts of interest.

Ethics Approval: Not applicable — this is a theoretical assessment involving no human participants, animal subjects, or sensitive data.

Consent to Participate: Not applicable.

Author Contributions: R.B.Q.-G. is the sole author and performed all aspects of the research, analysis, and writing.

Data Availability: All source artifacts are archived at the GitHub repository `QNFO/acrp07-qubit-delusion-val` and on Zenodo at the DOI listed in the frontmatter. The Qubit Delusion series papers are available at DOIs 10.5281/zenodo.21254143, 10.5281/zenodo.21254901, and 10.5281/zenodo.21484345.

Code Availability: Not applicable — this paper contains no original code.

Materials Availability: Not applicable.

Use of Artificial Intelligence: A large language model was used as a research assistant for literature synthesis and structured analysis under the supervision and editorial direction of the human author, who is solely responsible for all claims, judgments, and conclusions.

References

[1] QNFO Research Collective, “The Qubit Delusion: How Particle Ontology Sabotaged Quantum Computing,” Zenodo, 2026. DOI: 10.5281/zenodo.21254143.

- [2] R. B. Quni-Gudzinis, “Primitive Ultrametric Kernels: Valuation-Theoretic Classification of QEC Constructions” (ACRP-01 Consilient Synthesis), Zenodo, v2.2, 2026. DOI: 10.5281/zenodo.21727314.
- [3] QNFO Research Collective, “Beyond the Qubit: Constructive Paradigms for Post-Particle Computation,” Zenodo, 2026. DOI: 10.5281/zenodo.21254901.
- [4] R. B. Quni-Gudzinis, “The Adelic Cross-Domain Program: From the Fine-Structure Constant to the Standard Model,” Zenodo, v4.0, 2026. DOI: 10.5281/zenodo.21736300.
- [5] R. B. Quni-Gudzinis, “When Will Non-Archimedean Geometry Displace the Real Numbers? A Structured Assessment of the Adelic Substrate Thesis” (ACRP-08), Zenodo, v1.0, 2026. DOI: 10.5281/zenodo.21747228.